

# MANTLE ECOSYSTEM

AGENTIC PORTAL - v5

Intent-Driven Cross-Chain Onboarding · Dual-Interface UX · LayerZero OFT  
6-Source Data Waterfall · X + Discord Social Intelligence  
**NEW v5: Privy Social Auth · Embedded Wallets · Zero-Friction Web2→Web3 Onboarding**  
**Auto Network Switch — No Warning C, No Manual Steps, No Extra Screens**

|                  |          |          |
|------------------|----------|----------|
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# 1. PROJECT VISION & PHILOSOPHY

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The Mantle Agentic Portal exists to make every person on earth able to access Mantle DeFi — regardless of crypto experience. v5 achieves the deepest onboarding simplification yet: **the portal never asks a user to change their network**. It does it for them, silently, the moment a wallet connects.

**Intent over instruction:** Natural language → agent routing, bridging, approvals, execution. Users state goals, not steps.

**Zero crypto prerequisite:** Social login → embedded wallet auto-created. No MetaMask, no seed phrases, no extensions required.

**Network is infrastructure:** Switching to Mantle happens silently in a background hook. It is never surfaced as a user decision or warning.

**Guided funding only when needed:** If wallet is empty, portal shows funding options. If wallet is on the wrong chain, it self-corrects without user involvement.

**Chain-agnostic entry:** Solana, EVM, Bitcoin wallets all supported via Privy. Non-EVM users are guided to add an EVM wallet — that is the only chain-related prompt.

**Gasless first experience:** Paymaster covers first transactions. Empty-wallet users can still enter and explore.

## 2. WHAT WE FOUND ON [mantle.xyz/ecosystem](https://www.mantle.xyz/ecosystem)

Scraped from <https://www.mantle.xyz/ecosystem> — the authoritative dApp registry.

| Metric            | Value                                                                         |
|-------------------|-------------------------------------------------------------------------------|
| Total dApps       | 242                                                                           |
| Featured projects | 26                                                                            |
| Category filters  | 16                                                                            |
| Official X        | <a href="https://twitter.com/Mantle_Official">x.com/Mantle_Official</a>       |
| Official Discord  | <a href="https://discord.com/invite/0xMantle">discord.com/invite/0xMantle</a> |

### Category Counts

| Category                     | Count | Portal ID |
|------------------------------|-------|-----------|
| DEX & AMM (incl. Aggregator) | 56+   | dex       |
| Dev Tool / Infrastructure    | 29+   | infra     |
| Lending + Money Market       | 24    | lending   |
| Wallet / Custody             | 16    | wallet    |
| Bridge & Cross-Chain         | 12    | bridge    |
| Games & NFT                  | 12+   | gaming    |
| Vault / Yield                | 10    | yield     |
| Oracle                       | 6     | oracle    |
| RWA / Synthetic              | 5+    | rwa       |
| DAO + Social                 | 7+    | dao       |
| On-Ramp / Fiat               | 3     | fiat      |

### 3. FULL 242-PROJECT REGISTRY — FEATURED PROJECTS

| Project       | Category        | URL                | Badge                       |
|---------------|-----------------|--------------------|-----------------------------|
| INIT Capital  | Lending         | init.capital       | Featured, EcoFund           |
| Merchant Moe  | DEX             | merchantmoe.com    | Featured, EcoFund           |
| Vertex        | Derivatives     | vertexprotocol.com | Featured, Official          |
| mETH Protocol | Liquid Staking  | meth.mantle.xyz    | Featured                    |
| ONDO Finance  | RWA             | ondo.finance       | Featured                    |
| Ethena        | RWA / Synthetic | ethena.fi          | Featured, EcoFund           |
| AGORA (AUDS)  | Stablecoins     | agora.finance      | Featured, EcoFund, Official |
| IntentX       | Derivatives     | intentx.io         | Featured, EcoFund           |
| Stargate      | Bridge          | stargate.finance   | Featured                    |
| Pendle        | Yield           | pendle.finance     | Featured                    |
| Arkham        | Analytics       | intel.arkm.com     | Featured, Official          |
| TsunamiX      | Perps           | tsunami.finance    | Featured, EcoFund           |
| MethLab       | Lending         | methlab.xyz        | Featured                    |
| funny.money   | DEX             | funny.money        | Featured, Official          |
| Layer3        | Social          | layer3.xyz         | Featured, Official          |
| Galxe         | Social          | galxe.com          | Featured, Official          |
| Hyperlane     | Bridge/Dev      | hyperlane.xyz      | Featured, Official          |
| FBTC          | BTC Asset       | fbtc.com           | Featured                    |
| Catizen       | Games           | catizen.ai         | Featured, EcoFund           |
| MetaCene      | Games           | metacene.io        | Featured, EcoFund           |

## 4. ECOSYSTEM CATEGORY ARCHITECTURE

| Portal Category | mantle.xyz Labels               | Icon | Approx Count |
|-----------------|---------------------------------|------|--------------|
| All             | All                             | ■    | 242+         |
| DEX & AMM       | DEX, Aggregator, RFQ/Intents    | ■    | 58+          |
| Lending         | Lending, Money Market           | ■    | 24           |
| Derivatives     | Perpetual Futures, Intents      | ■    | 10+          |
| AI Agents       | (portal-exclusive)              | ■    | Growing      |
| Liquid Staking  | Restaking, LST                  | ■    | 3+           |
| Yield & Vaults  | Vault                           | ■    | 10           |
| Bridges         | Bridge                          | ■    | 12           |
| RWA & Stables   | RWA/Synthetic                   | ■    | 8+           |
| Games & NFT     | Games, NFT Marketplace          | ■    | 12+          |
| Infrastructure  | Oracle, Node Provider, Dev Tool | ■■   | 40+          |
| Social & Quests | Social, DAO Tool                | ■■   | 10+          |

## 5. PRIVY INTEGRATION — FULL ARCHITECTURE

### 5.1 What Privy Is and Why It Was Chosen

**Privy** is an embedded wallet and authentication SDK — acquired by Stripe in June 2025. It lets any user sign in with Google, Apple, Discord, email, or an existing wallet, and instantly receives a non-custodial EVM (and/or Solana) wallet — no seed phrases, no browser extensions, no friction. 100M+ wallets deployed across 1,500+ apps.

**No seed phrases ever:** 2-of-3 Shamir Secret Sharing. User never sees a private key.

**Auto wallet on login:** `createOnLogin`: 'users-without-wallets' provisions EVM + Solana wallets on first sign-in. Completely invisible to the user.

**15+ auth methods:** Google, Apple, Twitter, Discord, email, SMS, passkey, MetaMask, Phantom, WalletConnect, Coinbase, Farcaster, Telegram.

**EVM + Solana in one SDK:** Both chain types handled, with cross-chain account linking.

**Built-in funding UI:** `useFundWallet()` opens a native on-ramp modal. Supplemented by direct on-ramp partner links.

**Stripe acquisition:** Payment-rail integration with Stripe Checkout expected — card-to-DeFi in the most familiar checkout UI on earth.

### 5.2 Privy Security Model

| Share          | Location                             | Rule                                     |
|----------------|--------------------------------------|------------------------------------------|
| Device share   | User's browser (localStorage)        | Never leaves the browser unencrypted.    |
| Privy share    | Privy TEE infrastructure             | Privy cannot read it — even internally.  |
| Recovery share | User passkey / iCloud / hardware key | Used if device is lost. User-controlled. |

Any 2-of-3 reconstruct the key. Non-custodial: the portal and Privy cannot sign without the user.

### 5.3 All 15+ Supported Auth Methods

| Method          | Type            | Privy Key |
|-----------------|-----------------|-----------|
| Google          | Social OAuth    | 'google'  |
| Apple           | Social OAuth    | 'apple'   |
| Twitter / X     | Social OAuth    | 'twitter' |
| Discord         | Social OAuth    | 'discord' |
| GitHub          | Social OAuth    | 'github'  |
| Email + OTP     | Email           | 'email'   |
| SMS / Phone     | Phone           | 'sms'     |
| Passkey         | Biometric       | 'passkey' |
| MetaMask        | External EVM    | 'wallet'  |
| WalletConnect   | External EVM    | 'wallet'  |
| Coinbase Wallet | External EVM    | 'wallet'  |
| Phantom         | External Solana | 'wallet'  |

|           |                 |             |
|-----------|-----------------|-------------|
| Backpack  | External Solana | 'wallet'    |
| Farcaster | Web3 social     | 'farcaster' |
| Telegram  | Messaging       | 'telegram'  |



## 5.4 PrivyProvider Configuration for Mantle

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# 6. ONBOARDING STATE MACHINE — 3 SCREENS, NO NETWORK SCREEN

## 6.1 Design Principle: Network Switching Is Infrastructure, Not UX

This is the most important UX decision in the entire onboarding system. **Network switching must never be a user decision, a warning, a CTA, or a screen.** It is an invisible technical operation that the portal performs automatically the instant a wallet is connected.

The user's mental model is: 'I connected my wallet → I am on Mantle.' They should never know or care what chain their wallet was previously on. The portal absorbs that complexity entirely.

Analogy: when you connect to a new Wi-Fi network, your phone doesn't ask you 'are you sure you want to change networks?' and show a warning banner. It just connects. The Mantle portal treats network switching the same way.

## 3-Screen State Machine (network switch removed as screen)

| Screen           | State Condition          | User Sees                              | Triggered By               |
|------------------|--------------------------|----------------------------------------|----------------------------|
| 0 — Welcome      | Not authenticated        | Hero, value props, Get Started CTA     | User arrives               |
| 1 — Login        | Not authenticated        | 7 social login buttons                 | Tap Get Started            |
| 2 — Wallet Check | authenticated === true   | Wallet display, warnings A/B if needed | Privy auth completes       |
| Portal           | EVM wallet + no blockers | Full agentic portal                    | Tap Enter Portal           |
| (background)     | Any EVM wallet connects  | Nothing (silent)                       | useAutoSwitchToMantle hook |

## 6.2 Screen 0 — Welcome

Web2-friendly. No crypto terminology. Three value props: no seed phrases, gas covered, any chain. Single CTA: 'Get Started'. The word 'wallet' does not appear on this screen.

## 6.3 Screen 1 — Social Login

7 auth method buttons, Google first (highest Web2 conversion rate). Each calls login({ loginMethods: [method] }). On success, Privy fires authenticated = true, which triggers the useEffect to advance to Screen 2. Embedded EVM and Solana wallets are created in the background during this same auth flow.

## 6.4 Screen 2 — Wallet + Funding Check

The only user-facing wallet screen. By the time this renders, useAutoSwitchToMantle has already fired (or is firing). The screen shows: the user's wallet(s), a subtle 'Connecting to Mantle...' spinner if the switch is in progress (not a warning — just ambient feedback), Warning A if Solana-only, Warning B if balance is zero. The Enter Portal button is available as soon as an EVM wallet exists.

## 7. useAutoSwitchToMantle — THE CORE HOOK

### 7.1 How It Works

This hook is the single mechanism responsible for keeping every connected EVM wallet on Mantle at all times. It runs as a side effect, not as a UI flow:

**Trigger:** Fires in a `useEffect` whenever the `wallets` array changes — specifically when any new wallet address appears.

**Per-wallet tracking:** A `useRef` Set tracks which wallet addresses have already been switched. This prevents duplicate `switchChain` calls on re-renders.

**Silent chain read:** Calls `wallet.getChainId()` first. If already on Mantle (5000), the wallet is logged as switched and the hook exits. No unnecessary RPC calls.

**Silent switch:** Calls `wallet.switchChain(5000)` with no UI notification. Privy's embedded wallet switches instantly. External wallets (MetaMask, Coinbase) show their own native confirmation — this cannot be avoided for external wallets, but it is the wallet's own UI, not the portal's.

**Fallback on failure:** If `switchChain` fails (e.g. hardware wallet, some restricted external wallets), the error is logged to console only. The user sees nothing. The portal still opens — they may need to switch manually inside Interface 2 if they hit a tx that requires it.

**Ambient status:** The switching boolean returned by the hook drives a small inline 'Connecting to Mantle...' spinner in the wallet check screen. It is not a warning, not a blocker, not a CTA.

### 7.2 Full Hook Implementation

```
function useAutoSwitchToMantle(wallets, onSwitched) { const switched = useRef(new Set()); //
addresses already handled const [switching, setSwitching] = useState(false); useEffect(() =>
{ const evmWallets = wallets.filter(w => w.chainType === 'ethereum'); if
(!evmWallets.length) return; const runSwitch = async () => { for (const wallet of
evmWallets) { if (switched.current.has(wallet.address)) continue; // already done try {
const currentChain = await wallet.getChainId?().catch(() => null); if (currentChain ===
MANTLE_CHAIN_ID) { // Already on Mantle – mark and move on
switched.current.add(wallet.address); continue; } // Different chain – switch silently, no
UI setSwitching(true); await wallet.switchChain(MANTLE_CHAIN_ID);
switched.current.add(wallet.address); if (onSwitched) onSwitched(); } catch (err) { //
Silent fail – log only, never surface to user console.warn('[Mantle Portal] Auto-switch
failed for', wallet.address, '-', err.message); // Note: for external wallets (MetaMask
etc.), Privy's switchChain() // calls wallet_switchEthereumChain on the extension. If the
user // dismisses the MetaMask popup, this catch fires. We do not retry // or warn – the
user's action (dismiss) is respected silently. } finally { setSwitching(false); } } };
runSwitch(); // Dependency: re-run when wallet addresses change }, [wallets.map(w =>
w.address).join(',')]); return { switching }; }
```

### 7.3 Edge Cases and Fallbacks

| Scenario                             | Hook Behaviour                                | User Sees                           |
|--------------------------------------|-----------------------------------------------|-------------------------------------|
| Wallet already on Mantle             | <code>getChainId()</code> returns 5000 → skip | Nothing                             |
| Embedded EVM wallet (Privy-switched) | <code>switchChain()</code> succeeds instantly | Tiny spinner for <1s                |
| MetaMask — user accepts switch       | MetaMask popup → user confirms → done         | MetaMask native popup (unavoidable) |

|                                                    |                                                 |                         |
|----------------------------------------------------|-------------------------------------------------|-------------------------|
| MetaMask — user dismisses popup                    | popup fires → logged silently → no retry        | Nothing (silent)        |
| Hardware wallet (Ledger etc.) switchChain may fail | → logged silently                               | Nothing                 |
| Wallet connect fails entirely                      | Hook exits silently, portal still opens         | Nothing                 |
| User connects second wallet later                  | lets dep changes → hook re-runs for new address | Shows spinner if needed |

## 8. WARNING SYSTEM — ONLY 2 WARNINGS

### 8.1 Warning A: No EVM Wallet (Solana-only user)

Trigger: user has  $\geq 1$  Solana wallet and 0 EVM wallets.

```
const hasOnlySolana = solWallets.length > 0 && evmWallets.length === 0; // Shows only when
hasOnlySolana === true // Action button calls Privy's linkWallet() to add an EVM wallet. //
Privy auto-creates an embedded EVM wallet if user has none. const { linkWallet } =
usePrivy();
```

Message: "■ ■ ■ You also need an EVM wallet to access Mantle. You're connected with a Solana wallet. Mantle is EVM-compatible. Tap below to add one — it takes one tap. We'll bridge your Solana assets automatically once you're set up."

### 8.2 Warning B: Empty Wallet — You Need Assets

Trigger: user has  $\geq 1$  EVM wallet on Mantle, but its MNT balance is 0.

```
// Balance polling — runs every 8 seconds so on-ramp reflects immediately const mantleClient
= createPublicClient({ chain: mantleChain, transport: http(RPC) }); const balance = await
mantleClient.getBalance({ address: evmWallet.address }); const isEmpty = balance === 0n;
```

Message: "■ ■ ■ You need assets to begin using Mantle. Your wallet is ready and on Mantle, but has no balance. Fund it to start using protocols — under 5 minutes. You can also skip and explore the portal first."

### 8.3 What Was Removed — No Warning C

Previous versions included a 'Warning C' for wrong network, a 'Network Detection' screen (Screen 3), and a manual 'Switch to Mantle' CTA button. **All of these have been removed.**

Why they were wrong:

**Wrong network warning:** Puts a technical infrastructure problem on the user's plate. They don't know what 'Arbitrum' or 'chain ID 5000' means. They shouldn't need to.

**Network Detection screen:** An entire screen just to switch chains is extreme friction. It makes the portal feel like every other DeFi app that constantly asks you to switch networks.

**Manual 'Switch to Mantle' button:** A CTA for something the portal can and should do automatically. Asking the user to tap a button for a background operation is false friction.

**The replacement:** `useAutoSwitchToMantle()` — a silent background hook that handles the switch before Screen 2 even finishes rendering in most cases.

The only reason a network warning could ever be justified is if `switchChain()` fails and the portal genuinely cannot proceed without user help. Even then, the correct response is a *contextual* message at the point of transaction failure, not a blocking warning banner at onboarding entry.

### 8.4 Funding Options (4 On-Ramps + Bridge + CEX)

Shown inside Warning B's funding drawer when the user taps 'Fund My Wallet':

| Provider | URL                | Time    | Fee  | Best For                                              |
|----------|--------------------|---------|------|-------------------------------------------------------|
| Transak  | global.transak.com | < 5 min | 1–2% | Card + bank, 100+ countries, pre-fills wallet address |
| Banxa    | banxa.com          | < 5 min | 1–3% | 130+ fiat currencies, global coverage                 |

|              |                |         |      |                                     |
|--------------|----------------|---------|------|-------------------------------------|
| Onramp.money | onramp.money   | < 1 min | ~1%  | Fastest conversion, minimal steps   |
| Alchemy Pay  | alchemypay.org | < 3 min | 1–2% | Global gateway, 50+ fiat currencies |

Additionally: bridge-from-chain text card (agent handles intent) and CEX copy-address card (Binance / OKX / Bybit withdrawal). Transak URL includes {walletAddress} as a query param so the address is pre-filled — one fewer step for the user.

## 9. PRIVY HOOK REFERENCE

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### 9.1 usePrivy

```
const { ready, // boolean – SDK initialised (always wait for this) authenticated, // boolean
  – user is logged in user, // User object – id, email, google, twitter, wallet,
  linkedAccounts login, // fn(options?) – open login modal logout, // fn() – sign user out
  linkWallet, // fn() – add another wallet to user account createWallet, // fn() – manually
  create embedded wallet exportWallet, // fn() – let user export their private key } =
  usePrivy();
```

### 9.2 useWallets

```
const { wallets } = useWallets(); // wallet.address – public address // wallet.chainType –
'ethereum' | 'solana' // wallet.walletClientType – 'privy' (embedded) | 'metamask' |
'coinbase_wallet' | ... // wallet.getChainId() – current chain ID (async) //
wallet.switchChain(id) – switch to chain (async, used by auto-switch hook) //
wallet.getEthersProvider() – ethers.js provider // wallet.getWalletClient() – viem wallet
client const evmWallets = wallets.filter(w => w.chainType === 'ethereum'); const solWallets
= wallets.filter(w => w.chainType === 'solana'); const embeddedEVM = wallets.find(w =>
w.walletClientType==='privy' && w.chainType==='ethereum');
```

### 9.3 useFundWallet

```
const { fundWallet } = useFundWallet(); await fundWallet(address, { chain: mantleChain,
amount: '50', // suggested USD (optional) asset: 'native-currency', }); // Opens Privy's
native funding modal: Coinbase Pay + MoonPay. // For Transak/Banxa/Onramp.money, open URLs
with wallet address param instead.
```

### 9.4 useEmbeddedWallet

```
import { useEmbeddedWallet, isNotCreated, isConnected } from '@privy-io/react-auth'; const {
state, createWallet } = useEmbeddedWallet(); // state.status: 'none' | 'not-created' |
'connecting' | 'connected' | 'error' if (isNotCreated(state)) await createWallet(); if
(isConnected(state)) console.log('Ready:', state.wallet.address);
```

## 10. COMPLETE COMPONENT CODE REFERENCE

The full component is in **privy-mantle-onboarding.jsx** (712 lines). Below is the structural summary and integration pattern.

```
// File: privy-mantle-onboarding.jsx – structural overview // // Constants: //
MANTLE_CHAIN_ID = 5000 // mantleChain = { id, name, nativeCurrency, rpcUrls, blockExplorers
} // mantleClient = viem publicClient pointing at rpc.mantle.xyz // ONRAMP_OPTIONS =
[Transak, Banxa, Onramp.money, Alchemy Pay] // AUTH_METHODS = [google, twitter, discord,
apple, email, sms, wallet] // // Core hook (THE KEY PIECE): //
useAutoSwitchToMantle(wallets, onSwitched) // → fires silently on wallet connect // →
switches to Mantle with no user interaction // → returns { switching } for ambient spinner
only // // Sub-components (presentational): // StepDots – 2-dot progress indicator //
WarningBanner – orange / blue warning card (used for A and B only) // WalletBadge – address
display + copy + green dot if on Mantle // NetworkStatusBar – tiny spinner shown while
switching (NOT a warning) // OnrampCard – on-ramp provider link card // // Screens (3
total): // WelcomeScreen – Web2 entry. No crypto terms. // LoginScreen – Social auth
buttons. Calls usePrivy().login() // WalletCheckScreen – Warning A, Warning B, funding
drawer, Enter Portal CTA // // Orchestrator: // MantlePrivyOnboarding – step state machine
(0/1/2), Privy hooks, auto-switch // // Root export: // MantlePrivyRoot – PrivyProvider
wrapper with full Mantle config
```

### App.jsx Integration

```
import { useState } from 'react'; import MantlePrivyRoot from
'./components/privy-mantle-onboarding'; import MantleAgenticPortal from
'./components/mantle-agentic-portal'; export default function App() { const [ready,
setReady] = useState(false); const [userWallets, setUserWallets] = useState([]); const
[privyUser, setPrivyUser] = useState(null); return ready ? : { setUserWallets(wallets);
setPrivyUser(user); setReady(true); }} /> }
```



# 11. NEW USER ONBOARDING — FULL CROSS-CHAIN FLOW

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## Step 1 — Visit portal

Web2-friendly hero. No crypto terms. 'Get Started' CTA.

## Step 2 — Social login via Privy

Google / Apple / Discord. EVM + Solana embedded wallets created in background.

## Step 3 — Auto-switch fires

useAutoSwitchToMantle() runs silently. Wallet moves to Mantle. User sees a small spinner at most.

## Step 4 — Wallet Check screen renders

Wallet shown on Mantle (green dot). Warning A if Solana-only. Warning B if empty.

## Step 5 — Fund wallet if needed

Card on-ramp, bridge intent via agent, or CEX withdrawal. Balance polling updates live.

## Step 6 — Enter Portal

One tap. No network screen. No network warning. Portal loads immediately.

## Step 7 — Drop bridge intent (if assets on other chain)

'Bridge \$100 from Solana to mETH'. Agent routes via LayerZero OFT. Paymaster covers gas.

## Step 8 — Explore ecosystem

Agent-guided discovery. DeFi Llama + X + Discord intelligence. 242 projects from mantle.xyz.

## Step 9 — Use dApp (Interface 2)

Embedded dApp, agent co-pilot, intent execution. Everything on Mantle.

# 12-13. LAYERZERO, PAYMASTER, DATA, SOCIAL, AGENT (REFERENCE)

| Component                | Technology                                      | Documented In     |
|--------------------------|-------------------------------------------------|-------------------|
| LayerZero OFT Bridge     | @layerzerolabs/lz-evm-sdk                       | v3 PDF Section 4  |
| ERC-4337 Paymaster       | @biconomy/account                               | v3 PDF Section 5  |
| Dual-Interface UX        | React state machine + iframe                    | v3 PDF Section 6  |
| DeFi Llama + 6 fallbacks | DeFiLlama→Dune→Graph→Nansen→Mobula→DeFiScreener | v4 PDF Section 10 |
| X (Twitter) Scraping     | X API v2 + Apify scraper                        | v4 PDF Section 11 |
| Discord Scraping         | discord.js + Redis                              | v4 PDF Section 12 |
| Claude Streaming Agent   | claude-sonnet-4-20250514, 8 tools               | v4 PDF Section 13 |

## 14. INSTALLATION & ANTIGRAVITY SETUP (v5)

### 14.1 Dependencies

| Package                      | Version | Purpose                                            |
|------------------------------|---------|----------------------------------------------------|
| @privy-io/react-auth         | ^3.14.1 | Social auth + embedded wallets (HTTPS required)    |
| react                        | ^18.3   | Core UI                                            |
| viem                         | ^2.x    | Mantle RPC client, balance checks, chain switching |
| lucide-react                 | ^0.383  | Icons (Loader, Shield, AlertTriangle, etc.)        |
| tailwindcss                  | ^3.4    | Utility CSS                                        |
| @layerzerolabs/lz-evm-sdk    | ^3.0    | OFT bridging                                       |
| @biconomy/account            | ^4.1    | ERC-4337 Paymaster                                 |
| wagmi                        | ^2.x    | Additional EVM utilities                           |
| @solana/wallet-adapter-react | ^0.15   | Solana external wallets                            |
| discord.js                   | ^14.x   | Discord community bot                              |
| ioredis                      | ^5.x    | Redis cache for Discord messages                   |
| @duneanalytics/client-sdk    | ^1.x    | Dune Analytics fallback                            |
| apify-client                 | ^2.x    | Apify Twitter scraper                              |

### 14.2 Environment Variables

```
# Privy (REQUIRED FIRST) REACT_APP_PRIVY_APP_ID=clxxxxxxxxxxxxxxxx # from dashboard.privy.io
# ■ HTTPS required – Privy's Web Crypto API silently breaks on plain http:// # localhost is
# exempt. All deployments must use HTTPS. REACT_APP_MANTLE_RPC=https://rpc.mantle.xyz
MANTLE_CHAIN_ID=5000 REACT_APP_WALLETCONNECT_PROJECT_ID=...
REACT_APP_ANTHROPIC_API_KEY=sk-ant-...
REACT_APP_BICONOMY_PAYMASTER_URL=https://paymaster.biconomy.io/api/v2/...
REACT_APP_LAYERZERO_APP_ID=... DUNE_API_KEY=... GRAPH_API_KEY=... NANSEN_API_KEY=...
MOBULA_API_KEY=... MESSARI_API_KEY=... TWITTER_BEARER_TOKEN=... APIFY_TOKEN=...
DISCORD_BOT_TOKEN=... REDIS_URL=redis://localhost:6379
```

### 14.3 Antigravity Steps

1. Create React workspace in Antigravity.
2. Paste **privy-mantle-onboarding.jsx** and **mantle-agentic-portal.jsx** into components/.
3. Update **App.jsx**: MantlePrivyRoot → onComplete → MantleAgenticPortal (Section 10 code).
4. Add all packages via Antigravity package manager.
5. **Critical:** Ensure HTTPS (or localhost). Privy silently breaks on http:// — no error message.
6. Set REACT\_APP\_PRIVY\_APP\_ID from dashboard.privy.io (free account).
7. Add Tailwind directives to index.css.
8. Run dev server. Test Google login → embedded wallet auto-creates.
9. Verify Screen 2: wallet shown → balance check runs → funding options appear if empty.
10. Verify no network warning anywhere — wallet should be on Mantle automatically.

**11.** Deploy to HTTPS before demoing Privy to stakeholders.

# 15. ENHANCEMENT ROADMAP

| Phase     | Milestone        | Features                                                                | Timeline |
|-----------|------------------|-------------------------------------------------------------------------|----------|
| 0 — Now   | Hackathon MVP    | Privy social auth, embedded wallet, auto-switch, warning A/B, mod       | Day 1-3  |
| 1 — Alpha | Live Auth + Data | Privy on HTTPS, DeFi Llama live, balance polling, on-ramp deep-link     | Day 4-7  |
| 2 — Beta  | Bridge + Funding | LayerZero OFT, Paymaster, Transak/Banxa, real mETH/USDY minting         | Week 2   |
| 3 — V1    | Full Portal      | Interface 2, X + Discord intelligence, all 242 projects from mantle.xyz | Week 3-4 |
| 4 — V2    | Scale + Stripe   | Stripe/Privy card-to-DeFi, smart account batching, portfolio tracking   | Month 2  |
| 5 — V3    | Autonomous       | Proactive agent alerts, mobile app (React Native + Privy)               | Month 3+ |

## 16. TROUBLESHOOTING & SUPPORT (v5)

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### 16.1 Auto-Switch Issues

#### Issue: Auto-switch doesn't fire on MetaMask connect

MetaMask external wallets require the user to confirm the chain switch in their extension popup. The hook calls `switchChain()` correctly, but MetaMask shows its own native popup. This is unavoidable for external wallets — it is the browser extension's security model. For embedded Privy wallets, the switch is fully silent and instant. Encourage users to use the embedded wallet (auto-created on social login) for the smoothest experience.

#### Issue: Auto-switch fires but wallet still shows wrong chain

The wallet address may already be in the switched Set from a previous session. Clear the Set reference by calling `switched.current.clear()` on logout, or key the Set on user ID rather than just wallet address.

#### Issue: switching stays true indefinitely

The `setSwitching(false)` call is in the finally block and should always fire. Check for an unhandled promise rejection that bypasses the finally. Add a safety timeout: `setTimeout(() => setSwitching(false), 5000)` as a fallback.

#### Issue: `getChainId()` returns null for new embedded wallet

Privy's embedded wallet may not expose `getChainId()` immediately after creation. Add `.catch(() => null)` as shown in the hook code and proceed to `switchChain()` directly — it is safe to call even if the current chain is unknown.

### 16.2 Privy Issues

#### Issue: Privy silently breaks — wallets don't work

You are on `http://` not `HTTPS`. Privy's Web Crypto API (key sharding) requires a secure context. `localhost` is exempt. All other hosts must use `HTTPS`. This produces no error message.

#### Issue: Embedded wallet not created after login

Check `createOnLogin`: 'users-without-wallets' in PrivyProvider config and in the Privy dashboard under your app's Embedded Wallets settings.

#### Issue: `useWallets()` returns `[]`

Wait for `ready === true` from `usePrivy()` before relying on wallet data.

#### Issue: WalletConnect not working

Privy does not include WalletConnect by default. Install with the WalletConnect connector and provide `REACT_APP_WALLETCONNECT_PROJECT_ID`.

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### 16.3 All Key Resources

- Privy Dashboard: <https://dashboard.privy.io>
- Privy React Quickstart: <https://docs.privy.io/basics/react/quickstart>
- Privy `useWallets`: <https://docs.privy.io/wallets/wallets/get-a-wallet/get-connected-wallet>

- Privy useFundWallet: <https://docs.privy.io/wallets/embedded/funding>
- Privy Solana: <https://privy.io/blog/solana-support>
- Transak: <https://transak.com> | Banxa: <https://banxa.com> | Onramp.money: <https://onramp.money>
- Mantle Ecosystem: <https://www.mantle.xyz/ecosystem> (242 projects)
- Mantle Discord: <https://discord.com/invite/0xMantle>
- LayerZero OFT: <https://docs.layerzero.network/v2/developers/evm/oft/quickstart>
- Biconomy Paymaster: <https://docs.biconomy.io/paymaster>
- Anthropic API: <https://docs.anthropic.com>
- DeFi Llama API: <https://defillama.com/docs/api>
- Dune Analytics: <https://docs.dune.com>
- The Graph: <https://thegraph.com/docs>
- Nansen: <https://docs.nansen.ai> | Mobula: <https://docs.app.mobula.io>
- X API v2: <https://developer.twitter.com/en/docs/twitter-api>
- discord.js: <https://discord.js.org>
- Mantle Docs: <https://docs.mantle.xyz>

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# READY TO BUILD

Any user with a Google account can be on Mantle in under 60 seconds. Network switching is invisible. Warnings are minimal. The agent does the rest.

| Component                                  | File                                       | Section   |
|--------------------------------------------|--------------------------------------------|-----------|
| PrivyProvider — full Mantle config         | privy-mantle-onboarding.jsx                | 5.4       |
| 15+ social auth methods                    | privy-mantle-onboarding.jsx                | 5.3 / 6.3 |
| Embedded wallet auto-creation              | PrivyProvider createOnLogin config         | 5.4       |
| useAutoSwitchToMantle hook (KEY)           | privy-mantle-onboarding.jsx                | 7         |
| Silent auto-switch on wallet connect       | Hook fires before Screen 2                 | 7.1-7.2   |
| No Warning C — network is infrastructure   | By design — auto-switch replaces it        | 8.3       |
| Warning A: No EVM wallet                   | WalletCheckScreen                          | 8.1       |
| Warning B: Empty wallet + funding UI       | WalletCheckScreen                          | 8.2 / 8.4 |
| 4 on-ramp options (pre-filled address)     | OnrampCard components                      | 8.4       |
| 3-screen state machine (no network screen) | MantlePrivyOnboarding                      | 6         |
| App.jsx integration pattern                | Section 10                                 | 10        |
| 242-project registry from mantle.xyz       | Seed data source confirmed                 | 3         |
| 6-source data waterfall                    | DeFiLlama→Dune→Graph→Nansen→Mobula→Messari | 12        |
| X + Discord social intelligence            | Documented in v4                           | 12        |
| Claude streaming agent (8 tools)           | claude-sonnet-4-20250514                   | 12        |